

Figure 1: Traditional network architecture vs SDN architecture

ofswitch13 library is used. All the commands given below have been tested on Ubuntu 16.04 and might change for other versions or distributions.

Before that, there are a few bundles to be introduced on the system:

```
$ sudo apt-get install build-essential gcc g++ python git
mercurial unzip cmake
$ sudo apt-get install libpcap-dev libxerces-c-dev libpcr3-dev
flex bison
$ sudo apt-get install pkg-config autoconf libtool libboost-dev
```

In order to utilise ofswitch13 as a static library, you need to introduce the Netbee library, as the ofswitch13 library code relies upon it.

```
$ wget https://bitbucket.org/ljerezchaves/ofswitch13-module/
downloads/nbeesrc.zip
$ unzip nbeesrc.zip (for unzipping)
$ cd netbee/src/
$ cmake .
$ make
$ sudo cp ../bin/libn*.so /usr/local/lib
$ sudo ldconfig
$ sudo cp -R ../include/* /usr/include/
```

Now, clone the repository of the ofswitch13 library, as follows:

```
$ git clone https://github.com/ljerezchaves/ofswitch13
$ cd ofswitch13
$ ./boot.sh
$ ./configure --enable-ns3-lib
$ make
```

Integrating OFSwitch with ns-3

To install ns-3.26, use the following command:

```
$ hg clone http://code.nsnam.org/ns-3.26
```

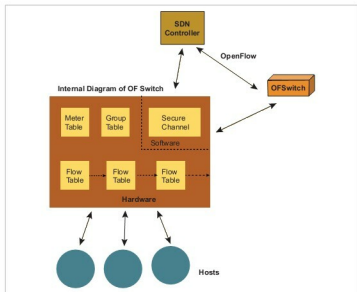


Figure 2: OpenFlow switch components

In the ns-3.26 directory, download the repository of OFSwitch13, as follows:

```
$ hg clone https://bitbucket.org/ljerezchaves/ofswitch13-
module src/ofswitch13
$ cd src/ofswitch13
$ hg update 3.1.0
$ cd ../..
$ patch -p1 < src/ofswitch13/utills/ofswitch13-src-3_26.patch
$ patch -p1 < src/ofswitch13/utills/ofswitch13-doc-3_26.patch
```

The file `ofswitch13-src-3_26.patch` will allow OFSwitch to get raw packets from nodes (devices). To do this, it will create a new OpenFlow receive callback at `CsmaNetDevice` and `virtualNetDevice`. The file `ofswitch13-doc-3_26.patch` is optional but preferable.

After successful installation, configure the module, as follows:

```
$ ./waf configure --with-ofswitch13=path/to/ofswitch13
$ ./waf configure --enable-examples --enable-tests
```

Now, we're all set. Just build the simulator using the following command:

```
$ ./waf
```

Enjoy the ns3.26 simulator with the power of SDN, i.e., OFSwitch 1.3.

Simulating the basic network topology with SDN based OFSwitch

In this section of the article, we'll simulate a basic network topology with three hosts, a switch and a controller.

Figure 3 demonstrates the topology of the network that we want to create. It includes three hosts, one switch and one controller.